GCSE Combined Science misconception bank

Biology, chemistry and physics. The mix-ups that survive into the exam unless you catch them.

These are hinge questions for mistakes that come up again and again at GCSE. The topics are the ones pupils meet on AQA and Edexcel courses. They are not past papers, and they are not a mark scheme. Named texts, case studies and optional units differ by school. Swap the example. Keep the mistake.

Each wrong answer is a reason to reteach. The shaded box is for you. Cover it if the page is on the board.

36 hinges.

PHOTOSYNTHESIS · 1

Where does a plant get the glucose it uses?

A. It makes glucose by photosynthesis.
B. It takes glucose up from the soil through its roots.
C. It takes in glucose, ready-made, from the air.
D. It gets glucose by eating insects.

Correct: A.

B. Roots were treated as if they absorb food. They take in water and mineral ions.

C. Carbon dioxide was confused with a ready supply of glucose.

D. A plant was given an animal's way of feeding.

Reteach: Photosynthesis builds glucose from carbon dioxide and water, using light energy. Mineral ions from the soil are not the plant's food.

RESPIRATION · 2

When does a plant respire?

A. All the time.
B. Only at night, when photosynthesis has stopped.
C. Only in the light.
D. Never. Photosynthesis replaces it.

Correct: A.

B. Respiration was switched off in the daytime to "make room" for photosynthesis.

C. Respiration was given the same condition as photosynthesis.

D. The two processes were treated as alternatives.

Reteach: Plant cells respire day and night. Photosynthesis needs light. In the light, both can be happening.

RESPIRATION · 3

Which sentence is accurate?

A. Breathing moves air. Respiration is the reaction in cells that releases energy.
B. Breathing and respiration are two words for the same thing.
C. Respiration only happens in the lungs.
D. Respiration makes oxygen.

Correct: A.

B. The movement of air was confused with the reaction in the cells.

C. The place where air is exchanged was treated as the place energy is released.

D. Respiration was swapped with photosynthesis. Aerobic respiration uses oxygen.

Reteach: Breathing (ventilation) moves air. Gas exchange happens at the alveoli. Respiration is the reaction in cells that releases energy from glucose.

CIRCULATION · 4

Which vessel carries deoxygenated blood?

A. The pulmonary artery.
B. No artery does. Every artery carries oxygenated blood.
C. The aorta.
D. The pulmonary vein.

Correct: A.

B. "Artery" was defined by oxygen, not by direction. Arteries leave the heart.

C. The aorta carries oxygenated blood from the heart to the body.

D. The pulmonary vein brings oxygenated blood from the lungs back to the heart.

Reteach: Arteries carry blood away from the heart. The pulmonary artery goes to the lungs and the blood in it is deoxygenated. Do not teach "artery means oxygen".

OSMOSIS · 5

Osmosis is the net movement of

A. water, from dilute to more concentrated, through a partially permeable membrane.
B. any particles, from concentrated to dilute, through any gap.
C. water, from concentrated to dilute, through a partially permeable membrane.
D. oxygen, by active transport, using energy.

Correct: A.

B. Osmosis was collapsed into diffusion, and the direction of the water was reversed.

C. The direction of the water was reversed. Water moves towards the more concentrated solution.

D. A different process was named. Osmosis is passive, and it is water.

Reteach: Osmosis is water only, through a partially permeable membrane, from a dilute solution to a more concentrated one. Diffusion can move other particles and does not need that membrane.

ENZYMES · 6

An enzyme is put in ice, then warmed back to room temperature. What is true?

A. The cold slowed it. Warming can restore the activity, because it was not denatured.
B. The cold denatured it, so the shape is permanently destroyed.
C. Cooling speeds an enzyme up.
D. Enzymes ignore temperature.

Correct: A.

B. Cold was treated as if it did the same damage as too much heat.

C. The temperature effect was reversed.

D. The rate was treated as fixed.

Reteach: Low temperature slows collisions. The shape is still there, so warming can restore activity. Denaturing is a change of shape from too much heat or the wrong pH, and that is not undone by cooling.

CELL DIVISION · 7

Which division makes gametes?

A. Meiosis. It produces four genetically different cells with half the chromosomes.
B. Mitosis. That is the division used to make sperm and eggs.
C. Mitosis, and the gametes are genetically identical to the parent cell.
D. Osmosis, because gametes have to move.

Correct: A.

B. Mitosis and meiosis were swapped.

C. The products of mitosis were given the job of gametes.

D. undefined

Reteach: Mitosis makes two genetically identical cells, for growth and repair. Meiosis makes gametes and halves the chromosome number.

INHERITANCE · 8

A dominant allele is one that

A. is expressed when only one copy is present.
B. is the allele most people have.
C. is always better for survival.
D. is only passed on by one sex.

Correct: A.

B. Dominant was used to mean common.

C. Dominant was used to mean stronger or healthier.

D. Dominance was confused with sex linkage.

Reteach: Dominant means the allele shows in the phenotype even if the other allele is different. It does not mean common, and it does not mean better.

EVOLUTION · 9

A person trains for years and builds a lot of muscle. Their child is born with ordinary muscles. Why does that fit what you teach?

A. The child inherits genes, not a change the parent built by training.
B. The child should inherit the extra muscle, because a useful change is passed on.
C. Exercise rewrites the parent's genes, and the child receives the new version.
D. Children inherit every change a parent makes to their body.

Correct: A.

B. A change made during one lifetime was treated as heritable. That is the giraffe-neck mistake.

C. Use was treated as if it edits the DNA that is passed on.

D. Acquired characteristics were treated as the normal mechanism.

Reteach: Offspring inherit genes that were already in the parents. A change from use, in one lifetime, is not passed on.

INFECTION · 10

A pupil has flu and asks for antibiotics. What is the accurate reply?

A. Antibiotics do not work on flu, because flu is caused by a virus.
B. Yes. Antibiotics kill the flu virus.
C. Flu is caused by bacteria, so a virus medicine is the wrong idea.
D. Antibiotics are a type of vaccine, so they prevent flu after you already have it.

Correct: A.

B. Antibiotics were treated as if they work on viruses.

C. The cause of flu was mislabelled. Flu is viral.

D. Two different ideas were merged, and a vaccine was turned into a cure.

Reteach: Antibiotics can work against some bacteria. They do not kill viruses. Flu is a viral infection.

INFECTION · 11

A vaccine protects you because it

A. shows the immune system an antigen in a safe form, so a later response can be faster.
B. cures the disease after you are already ill.
C. keeps you away from other people, and that is the whole of the effect.
D. destroys every microbe already living in your body.

Correct: A.

B. Vaccination was treated as a treatment for an illness you already have.

C. The immune response was left out.

D. A vaccine was described as a general poison for microbes.

Reteach: Vaccination lets the immune system prepare for a later infection. It is not a cure you take once the disease has already developed.

ECOLOGY · 12

In a food chain, the arrow

A. points to the organism that receives the energy. It means “is eaten by”.
B. points at the food, because it means “eats”.
C. points from the biggest organism to the smallest.
D. shows the path of blood.

Correct: A.

B. The arrow was drawn as a finger pointing at dinner.

C. Size was used instead of energy.

D. undefined

Reteach: Grass → rabbit means energy moves from the grass to the rabbit. The rabbit eats the grass. Read the arrow as “is eaten by”.

ATOMIC STRUCTURE · 13

Two isotopes of the same element have

A. the same number of protons and a different number of neutrons.
B. a different number of protons.
C. the same mass, and a different number of electrons. That is what “isotope” means.
D. no nucleus, which is why the mass can change.

Correct: A.

B. A different element was described. Proton number fixes the element.

C. Ions were described. Ions differ in electrons. Isotopes differ in neutrons.

D. undefined

Reteach: Same element means the same proton number. Isotopes differ in neutrons, so the mass number can differ. Ions differ in electrons.

ELECTRONIC STRUCTURE · 14

A sodium atom has 11 electrons. How are they arranged?

A. 2, 8, 1
B. 8, 3
C. 11
D. 2, 9

Correct: A.

B. The first shell was filled with 8. The first shell holds 2.

C. All the electrons were put in the first shell.

D. The second shell was overfilled. It holds up to 8 at this level.

Reteach: Fill the shells 2, then 8, then 8. Sodium is 2, 8, 1. That single outer electron is why a sodium atom forms a +1 ion.

BONDING · 15

How is sodium chloride held together?

A. Ionic bonding. An electron is transferred, then the oppositely charged ions attract.
B. Covalent bonding. The atoms share a pair of electrons.
C. Metallic bonding. A sea of electrons surrounds positive ions.
D. It is a mixture, so there is no bond.

Correct: A.

B. Sharing was used for a compound that is ionic.

C. The model for a metal was used for a salt.

D. A compound was called a mixture.

Reteach: Sharing electrons is covalent. A sea of delocalised electrons is metallic. Sodium chloride is ionic: Na^+ and Cl^- attract.

CONSERVATION OF MASS · 16

Wood burns on an open fire and the ash weighs less than the wood. A pupil says the missing mass was destroyed. What is the better account?

A. The atoms were not destroyed. Gases, including carbon dioxide, left up the chimney.
B. The flame destroyed that mass.
C. The ash should have gained mass, because fire adds new atoms.
D. Mass is only conserved for metals, not when wood burns.

Correct: A.

B. Burning was treated as if it deletes atoms.

C. Heat was treated as a substance that becomes part of the ash.

D. Conservation was limited to one type of reaction.

Reteach: Atoms are rearranged, not destroyed. In a sealed container the mass stays the same. On an open fire the container you weighed has lost gas.

ACIDS · 17

Which solution is more acidic?

A. pH 2, compared with pH 6.
B. pH 6, because 6 is a larger number.
C. They are the same, because both numbers are acids.
D. pH 2 is an alkali, because low numbers are the opposite of acid.

Correct: A.

B. A bigger pH number was read as a stronger acid.

C. The scale was ignored once the word “acid” appeared.

D. The scale was reversed.

Reteach: A lower pH is more acidic. One step on the pH scale is a ten-times change in hydrogen ion concentration. pH 2 is more acidic than pH 6, not weaker.

NEUTRALISATION · 18

A few drops of alkali are added to a beaker of acid. What happens to the pH?

A. It stays below 7 if plenty of acid is left.
B. It must become 7, because any alkali neutralises all of the acid.
C. It must become 14.
D. It cannot change, because an acid’s pH is fixed.

Correct: A.

B. A small amount of alkali was treated as a complete neutralisation.

C. Any alkali was treated as if it makes a strong alkali, whatever the quantity.

D. Amounts were ignored.

Reteach: pH 7, for a strong acid with a strong alkali, is when the amounts match. A few drops leave acid over, so the pH stays acidic.

RATES · 19

What is true of a catalyst?

A. It speeds the reaction up, it is still there at the end, and it lowers the activation energy.
B. It is used up, which is why the reaction is faster.
C. It raises the activation energy, so more collisions succeed.
D. It increases the overall energy change of the reaction.

Correct: A.

B. A catalyst was treated as a reactant that gets consumed.

C. The energy change was put the wrong way. A higher barrier would slow the reaction.

D. The catalyst was given the power to change how much energy the reaction transfers overall.

Reteach: A catalyst offers a route with a lower activation energy. It is not used up, and it does not change the overall energy change of the reaction.

RATES · 20

A reaction is heated and goes faster. Why?

A. Particles move faster, collide more often, and more of those collisions pass the activation energy.
B. The particles get larger, so they hit harder.
C. The activation energy increases, so more collisions count.
D. Heating means the particles collide less often.

Correct: A.

B. Heating was imagined as making the particles grow.

C. Heating was said to raise the barrier. It does not. More particles can get over the same barrier.

D. The collision idea was reversed.

Reteach: Heat makes the particles faster. Collisions are more frequent, and a greater proportion of them have energy above the activation energy. The activation energy itself does not go up.

ENERGY CHANGES · 21

An endothermic reaction is run in a beaker. The thermometer

A. falls, because the mixture takes energy from the surroundings.
B. rises, because “endo” means energy is given out.
C. stays still, because an energy change always cancels out.
D. rises, because every chemical reaction gives out heat.

Correct: A.

B. Endothermic and exothermic were swapped.

C. “Energy is conserved” was misread as “the temperature cannot change”.

D. All reactions were treated as exothermic.

Reteach: Endothermic: energy is taken in, so the mixture cools. Exothermic: energy is given out, so the surroundings warm. The word does not mean “the thermometer goes up”.

ORGANIC CHEMISTRY · 22

Cracking hydrocarbons

A. breaks long molecules into shorter, more useful ones.
B. joins short alkanes together to make long ones.
C. is another name for polymerisation.
D. turns hydrocarbons into metals.

Correct: A.

B. Cracking was swapped with the idea of building a long chain.

C. Splitting a long molecule was confused with joining many small ones.

D. undefined

Reteach: Cracking splits long hydrocarbons into shorter ones, including alkenes. Polymerisation joins many small molecules into a long chain. They are opposite jobs.

SEPARATING MIXTURES · 23

Seawater is poured through filter paper. What happens to the dissolved salt?

A. It passes through with the water. Filtering does not make the water fresh.
B. It is caught on the paper, so the water that drips through is fresh.
C. It evaporates as soon as it touches the paper.
D. It turns into sand, which is why the paper clogs.

Correct: A.

B. A dissolved substance was treated as if it were an insoluble solid.

C. Filtering was confused with evaporation.

D. A chemical change was invented.

Reteach: Filter paper holds back insoluble bits. Dissolved salt stays in the solution. Distillation, or another desalination method, separates the water from the salt.

THE PERIODIC TABLE · 24

Neon is unreactive because

A. its outer shell is full.
B. it has no electrons.
C. its outer shell is half full, so it cannot bond.
D. it is a metal, with a sea of electrons that will not move.

Correct: A.

B. A full shell was confused with an empty atom. Neon is 2, 8.

C. A half-full shell is not neon’s structure, and it is not why group 0 is unreactive.

D. A noble gas was given the structure of a metal.

Reteach: Neon’s outer shell is full (2, 8), which is a stable arrangement. It does have electrons. “No electrons” is not the reason.

FORCES · 25

An astronaut goes from Earth to the Moon. What changes?

A. Weight decreases, because gravity is weaker. Mass stays the same.
B. Mass decreases, and weight stays the same.
C. Both fall to zero, because there is no gravity on the Moon.
D. Mass, in newtons, falls. Weight, in kilograms, stays the same.

Correct: A.

B. Mass and weight were swapped. Mass is the amount of matter.

C. Weaker gravity was treated as no gravity.

D. The units were swapped, and so was the conclusion.

Reteach: Mass is in kilograms and does not change. Weight is the pull of gravity, in newtons, and it is smaller on the Moon. The Moon's gravity is weaker than Earth's. It is not zero.

FORCES · 26

A book is still on a table. Which statement is right?

A. Its weight and the normal force from the table are balanced, so the resultant force is zero.
B. There are no forces on it, because it is still.
C. The only force is its weight, pulling down.
D. A forward force keeps it still.

Correct: A.

B. "Resultant force zero" was shortened to "no forces".

C. The support from the table was left out, which would mean the book accelerates.

D. A force was invented to explain "no motion". Steady motion, including staying still, needs no forward force.

Reteach: Still, or steady speed in a straight line, means the resultant force is zero. The forces are there. They cancel. It does not mean there are no forces.

FORCES · 27

The book is still. Its weight and the normal force from the table are equal and opposite. Are they a Newton's third-law pair?

A. No. Both act on the book. The third-law partner of the normal force is the book pushing down on the table.
B. Yes. Any two forces that are equal and opposite are a third-law pair.
C. Yes, because one is the weight and one is the support, and those words always make a pair.
D. No, because a still object cannot be part of a third-law pair.

Correct: A.

B. "Equal and opposite" was treated as the whole of the third law. A third-law pair acts on two different objects.

C. The names were paired without asking which object each force acts on.

D. The law was limited to moving objects. Third-law pairs exist whether or not the book is still.

Reteach: Weight and the normal force both act on the book, so they are a balanced pair on one object, not a third-law pair. The partner of "table pushes up on book" is "book pushes down on table".

ELECTRICITY · 28

A lamp is in a series circuit. The current in the wire after the lamp is

A. the same as the current in the wire before the lamp.
B. smaller, because the lamp uses up current.
C. zero, because the lamp uses all of the current.
D. larger, because the lamp adds energy into the current.

Correct: A.

B. Charge was treated as a fuel that gets consumed.

C. The same mistake, taken to the end of the wire.

D. Energy transfer was confused with an increase in current.

Reteach: In a series circuit the current is the same all the way round. The lamp transfers energy. It does not use up charge.

ELECTRICITY · 29

Which line is right?

A. Current is the flow of charge, in amps. Potential difference is the energy per coulomb, in volts.
B. Potential difference is another name for current.
C. Resistance is measured in volts.
D. Current is measured in volts, and potential difference in amps.

Correct: A.

B. Volts and amps were treated as the same measurement.

C. The unit of potential difference was given to resistance. Resistance is in ohms.

D. The units were swapped.

Reteach: Keep the three apart. Current (amps) is the flow. Potential difference (volts) is energy transferred per coulomb. Resistance (ohms) opposes the current.

ELECTRICITY · 30

A second resistor is added in parallel with the first. The total resistance

A. decreases, because there is another path for the current.
B. increases, because there is more resistance in the circuit.
C. stays the same, because parallel components do not affect each other.
D. becomes zero.

Correct: A.

B. The series rule was used for a parallel circuit.

C. A second path was treated as if it were electrically invisible.

D. Another path was treated as a short circuit with no resistance at all.

Reteach: Adding a resistor in parallel adds a path, so the total resistance falls. Adding a resistor in series is what makes the total resistance rise.

PARTICLE MODEL · 31

In a solid, the particles

A. vibrate about fixed positions. They do not travel through the solid.
B. do not move at all.
C. move freely, as they do in a gas, but more slowly.
D. are not there. A solid is one continuous block with no particles.

Correct: A.

B. Vibration was denied. The particles are not still.

C. The solid was given the arrangement of a gas.

D. The particle model was dropped for solids.

Reteach: Solids keep their shape because the particles stay in a regular arrangement. Those particles vibrate. They do not wander through the solid the way gas particles travel.

CHANGES OF STATE · 32

A puddle dries up in the sun. The water

A. has become vapour in the air. It has not been destroyed.
B. has been destroyed by the sun.
C. has turned into a new element that is no longer water, because drying is a chemical reaction that deletes it.
D. has sunk to the centre of the Earth.

Correct: A.

B. A change of state was treated as matter disappearing.

C. Evaporation was redescribed as a chemical change that removes the water.

D. undefined

Reteach: Evaporation is a change of state. The particles leave the liquid and mix with the air as gas. The water is still water.

ENERGY · 33

A pupil says temperature tells you how much thermal energy an object has. Which reply is sound?

A. A tiny spark can be very hot and still store less energy than a large amount of warm water.
B. The hotter object always stores more thermal energy.
C. Temperature and thermal energy are two names for one measurement.
D. Only fuels have thermal energy. Hot water does not.

Correct: A.

B. Temperature was used as if it were the energy total, with mass left out.

C. The two quantities were identified.

D. The store was limited to fuels.

Reteach: Temperature is how hot something is. The thermal energy stored also depends on the mass and the material. A bath of warm water can store more than a small, very hot spark.

WAVES · 34

Which is right about space?

A. Light can travel through a vacuum. Sound cannot, because sound needs particles.
B. Sound travels through space, and light needs air to travel.
C. Neither light nor sound can cross a vacuum.
D. Both cross a vacuum in the same way, because both are waves.

Correct: A.

B. The two were swapped. You can see the Sun. You would not hear an explosion in open space.

C. Light was given sound's limitation.

D. "Both are waves" was treated as "both need the same medium".

Reteach: Sound is a vibration of particles, so it needs a medium. Light does not. Space is a vacuum: light from the Sun reaches us, and sound does not travel across it.

RADIATION · 35

Which of these is ionising in the same sense as a gamma ray?

A. An X-ray.
B. A radio wave.
C. The infrared beam from a TV remote.
D. A microwave.

Correct: A.

B. A low-energy electromagnetic wave was given gamma's effect.

C. A household infrared beam was treated as ionising.

D. Microwaves were grouped with gamma because both are invisible waves.

Reteach: Ionising radiation can knock electrons off atoms. Gamma rays and X-rays do this, and so does higher-energy ultraviolet. Radio, microwave and infrared do not. Do not treat every invisible wave as gamma.

ENERGY · 36

A bulb is lit by a cell. The energy

A. is transferred, including to the thermal store of the room. It is not destroyed.
B. is used up, and disappears once the light has been made.
C. is created inside the bulb.
D. stays in the wire. The bulb only borrows it and hands it all back.

Correct: A.

B. "Used up" was taken to mean the energy no longer exists.

C. The bulb was treated as a source of new energy, rather than a place where energy is transferred.

D. The heating and the light were left out of the account.

Reteach: Energy is not created or destroyed. The circuit transfers energy from the cell's chemical store. The bulb then transfers it as light and as heating of the surroundings.

Run one hinge in two minutes

The questions in this bank are multiple choice, so a class can show A, B, C or D on mini whiteboards. That is the fast version. You see the letter, and you reteach the line that goes with it.

If you want the sentence behind the letter, use Get It?.

1. Type the topic, or paste the question. Edit what comes back. It is your question.
2. Pupils join with the class code on a phone, a laptop, or a school device. A first name is enough. No account, and nothing to install.
3. They answer in their own words, typed or spoken. It is not multiple choice. The sentence shows the mistake more clearly than a letter does.
4. In about two minutes you get green, amber or red. The colour is guidance. It can be wrong, and you can change it.
5. Reteach the line from this bank before you move on.

Where phones are banned, do not demand a phone. Use school devices, or stay with the boards.

A first name and the answers are sent to an external AI. If school policy forbids that, do not use the tool with pupil answers. The boards still work.

Free during the beta. No card.

dotheygetit.app